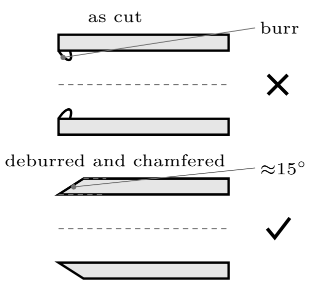
Craft 3: Deburr & Chamfer

Shop skill — transferable to anything you ever build

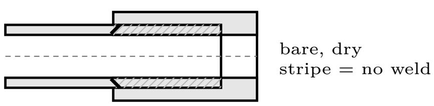
Two minutes that decide whether the joint is real

THE STANDARD

No burr anywhere, inside or out, and the outside edge of every pipe end carries a **small even chamfer** — roughly 1 to 2 mm at about 15 degrees, all the way round. Run a fingertip over the end: it should feel smooth in both directions and you should feel the chamfer as a gentle ramp, not a sharp step.



a burr scrapes the cement off as the pipe goes in



Why the outside burr matters It is a ridge of raised plastic exactly where the joint needs contact. As the pipe enters the socket the ridge acts like a scraper and wipes the cement off the leading surface, leaving a dry stripe down the joint. From outside the joint looks perfect.

Why the inside burr matters Inside a barrel it scores every projectile and disturbs the gas seal. Inside a chamber it collects residue. Neither is dramatic, and both are permanent.

Why the chamfer matters Without it, the leading edge of the pipe meets the socket taper as a sharp square corner and plows the cement ahead of it instead of sliding through it. The chamfer lets the pipe enter cleanly and keeps the cement where it belongs.

Two minutes This is the cheapest quality step in the whole build and the one most often skipped, because the pipe looks finished as soon as it is cut.

Tools A sharp knife works. A half-round file is better. A dedicated deburring tool is a few pounds and makes it trivial. All three are fine; doing nothing is not.

Method

1. **Inside first.** Run a knife blade or a rolled piece of sandpaper around the inner edge at a shallow angle. You are removing the raised lip, not cutting a bevel.
2. **Outside chamfer.** Hold a file or knife at about 15 degrees to the pipe surface and work all the way round, rotating the pipe rather than moving the tool in an arc. Rotating the work is what makes it even.
3. **Keep it small.** 1 to 2 mm. A large chamfer removes material the joint needs and can stop the pipe bottoming properly.
4. **Check by feel, then by eye.** Fingertip round the end, then hold it up and sight along it looking for a bright unchamfered patch.
5. **Wipe the swarf off.** Loose plastic shavings inside a chamber are exactly the debris you do not want near an igniter.

What each defect looks like later

If you skip	What you get
Outside deburr	A dry stripe in the weld you cannot see and cannot test. The joint holds — until the day it does not.
Chamfer	Cement scraped into a ring at the socket shoulder and starved along the joint face. Also a much harder push, which people mistake for a tight fit.
Inside deburr	Scored projectiles, a poorer gas seal, and a ledge that collects residue.
Cleaning up swarf	Loose plastic in the chamber, and grit in the cement on the next joint.

Check it yourself. Close your eyes and run a fingertip all the way round the inside edge and then the outside edge. Eyes closed genuinely helps — your fingertip resolves a burr far better than your eye does, and closing your eyes stops you believing what the pipe looks like instead of what it feels like.

The habit worth taking away

The defects that matter most are usually the ones that become invisible the moment you finish the next step. Once a pipe is in a socket you will never see that burr again — so the check has to happen *before* assembly, every time, on every end. Build the sequence so that inspection comes before the step that hides the thing being inspected.